

Fluoroquinolone Toxicity and FQAD

Evidence brief for prescribers - established mechanisms, clinical recognition, and the surveillance gap

Fluoroquinolone toxicity is easy to miss: symptoms may begin or worsen after the drug is stopped, cross several body systems, and appear despite a normal routine exam. The biological effects are not speculative: human-cell and mechanistic studies identify mitochondrial, neurologic, and connective-tissue effects that fit the clinical pattern.

1. Established Mechanisms of Fluoroquinolone Toxicity

Mitochondrial dysfunction and oxidative stress. In mammalian cells, fluoroquinolones drive mitochondrial dysfunction and reactive-oxygen-species overproduction, with oxidative damage to DNA, protein, and lipid (Kalghatgi et al., 2013).

Mitochondrial DNA disruption and direct human targets. Ciprofloxacin inhibits mitochondrial topoisomerase 2 (Top2 β), impairing mtDNA replication and transcription (Hansas et al., 2018). Chemical-proteomic mapping has also identified direct human off-targets of ciprofloxacin, including AIFM1 and IDH2 (Reinhardt et al., 2025).

Central nervous system effects. Fluoroquinolones antagonize GABA-A receptors and alter excitatory signaling (Halliwell et al., 1997). These effects provide a biological basis for recognized CNS adverse reactions, including seizures and neuropsychiatric symptoms.

Tendon and connective-tissue injury. In tendon cells and chondrocytes, fluoroquinolones generate oxidative stress and disturb collagen/matrix balance - consistent with the tendinopathy and tendon rupture that led to the 2008 boxed warning.

The clinical point. Fluoroquinolones can affect several human biological pathways, which helps explain why injury may present across multiple systems rather than as a single-organ reaction. **Some adverse effects may appear or progress after the drug is stopped.** Published reports document delayed onset extending weeks to months after exposure, while disabling effects may persist much longer.

2. What Remains Unknown

Several mechanisms of fluoroquinolone injury are well described. What is not yet established is why some patients recover after the drug is discontinued while others develop prolonged, delayed and potentially irreversible disability.

Persistent injury after fluoroquinolone exposure is clinically documented, but the biological processes responsible for that persistence - and the factors that determine who recovers and who does not - remain important research questions.

That distinction matters: the adverse effects are established; the full explanation for their duration is not.

3. Why FQAD Is Missed

FQAD is not a diagnosis of exclusion or a psychosomatic label. It is a serious, disabling multisystem condition with published clinical classification and diagnostic criteria. It is often missed because symptoms can appear after the drug is stopped, affect several parts of the body, and routine testing may be normal. If no one connects those symptoms back to the fluoroquinolone, the patient is often given another diagnosis instead. This is why informed consent, recognition, and longitudinal tracking matter.

A 2015 *BMJ Case Reports* series described four previously healthy, employed adults who developed a severe, disabling multisystem syndrome during fluoroquinolone use that progressed after discontinuation — including tendinopathy, muscle weakness, peripheral neuropathy, autonomic dysfunction, sleep disturbance, cognitive dysfunction, and psychiatric symptoms (Golomb et al., 2015). Formal diagnostic criteria have since been published (Pieper, Springer, 2nd ed., 2026), and fluoroquinolone adverse effects are codeable in ICD-10-CM (T36.AX5A / -D / -S).

4. The Surveillance Gap

Recognition depends on a system that can see. Spontaneous reporting (FAERS) captures only a fraction of events and is poorly suited to reactions that may begin after a short course. There is no longitudinal cohort or natural-history study to establish recovery trajectories, relapse rates, or the factors that separate those who recover from those who remain disabled. There is also no established patient registry. Without longitudinal follow-up, the true burden and clinical course remain poorly measured.

5. Informed Consent as the First Step in Recognition

Informed consent is where recognition begins. A patient who knows what serious reactions to watch for can connect a new tendon, nerve, neurologic, or multisystem symptom to a recent fluoroquinolone exposure, seek care, tell the next clinician, and report the adverse event. A patient who was never told may have no reason to make that connection.

That missing connection affects surveillance. If patients and clinicians do not recognize the reaction, it is less likely to be documented, reported, or followed over time. Informed consent is therefore not simply a warning at the moment of prescribing; it is the first step in recognition and pharmacovigilance.

The clinical ask is small: a brief, specific conversation at the point of prescribing. FDA Citizen Petition Docket **FDA-2026-P-5116** requests standardized informed-consent language so patients know what to watch for before exposure.

References

1. Kalghatgi S, Spina CS, Costello JC, et al. Bactericidal antibiotics induce mitochondrial dysfunction and oxidative damage in mammalian cells. *Sci Transl Med*. 2013;5(192):192ra85.
2. Hangaas A, Aasumets K, Kekäläinen NJ, et al. Ciprofloxacin impairs mitochondrial DNA replication initiation through inhibition of Topoisomerase 2. *Nucleic Acids Res*. 2018;46(18):9625-9636.
3. Reinhardt T, El Harraoui Y, Rothemann A, et al. Chemical proteomics reveals human off-targets of fluoroquinolone-induced mitochondrial toxicity. *Angew Chem Int Ed Engl*. 2025;64(18):e202421424.
4. Halliwell RF, Davey PG, Lambert JJ. Selective antagonism of the GABA-A receptor by ciprofloxacin and biphenylacetic acid. *Br J Pharmacol*. 1997;122:584-590.
5. Golomb BA, Koslik HJ, Redd AJ. Fluoroquinolone-induced serious, persistent, multisymptom adverse effects. *BMJ Case Rep*. 2015;2015:bcr2015209821.
6. Pieper S. Fluoroquinolone-Associated Disability FQAD: Side-effects of Fluoroquinolones. 2nd ed. Springer; 2026.
7. U.S. Food and Drug Administration. Drug Safety Communications on fluoroquinolones, 2008-2018.

Learn more

Fluoroquinolone Toxicity Study, NFP (fq100.org) - cellular- and molecular-level research into mechanisms of fluoroquinolone injury. DIMD Initiative (druginducedmito.org) - systems-level work on drug-induced mitochondrial dysfunction, FQAD recognition, longitudinal tracking, informed consent, and pharmacovigilance.

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